Sohan Jayram Reddy Kadiri

Seattle | sohanjreddy@outlook.com | (206) 380-6390 | github.com/sohanreddyk | linkedin.com/in/sohanreddyk | www.sohanreddy.org/

PROFESSIONAL SUMMARY

Data Scientist with experience building predictive models, forecasting systems and data-driven products using Python and SQL. Skilled in feature engineering, statistical evaluation, experimentation and model explainability, translating complex datasets into measurable business and operational decisions.

EDUCATION

Northeastern University Sep 2025 - May 2027
Master of Science, *Computer Science* GPA: 3.9

Siddaganga Institute of Technology Dec 2021 - Jun 2025
Bachelor of Technology, *Artificial Intelligence and Data Science*

PROJECTS

BIXI — Bike-Share Demand Forecasting Platform | Python, LightGBM, Optuna, Pandas, MLflow, SHAP, LIME, AWS

- Built a **15-minute demand forecasting pipeline across 1,100+ BIXI stations**, integrating historical trips, weather and temporal data to predict station-level arrivals and departures.
- Engineered **leakage-safe temporal and historical features** and tuned LightGBM with Optuna, achieving test RMSE of **1.035 for departures and 1.026 for arrivals**.
- Interpreted model behavior using **SHAP and LIME**, identifying historical demand and cyclical time features as key predictors and translating forecasts into station-rebalancing recommendations.

Fraud Detection System — Real-Time Fraud Analytics Platform | Python, XGBoost, Random Forest, Scikit-learn, Kafka, MLflow, Redis

- Built a **real-time fraud detection pipeline across 100K+ transactions**, engineering behavioral features for transaction velocity, geographic activity and amount deviation to quantify fraud risk.
- Trained and compared **Logistic Regression, Random Forest and XGBoost** across 25+ MLflow-tracked experiments, selecting the highest-performing model using ROC-AUC.
- Designed an **80/15/5 champion-challenger-shadow evaluation strategy** with drift monitoring, reducing false-positive rate by 18% while maintaining 95%+ fraud recall.

Predictive Maintenance ML System | Python, Scikit-learn, HistGradientBoosting, MLflow, SHAP, SciPy

- Developed a **predictive-maintenance classification pipeline on 10,000 sensor records**, benchmarking 3 models using a leakage-free 70/15/15 train-validation-test design.
- Selected and threshold-tuned **HistGradientBoosting using validation PR-AUC**, achieving **0.982 ROC-AUC, 0.909 PR-AUC and 88.2% failure recall** on the untouched test set.
- Implemented **SHAP explainability and Kolmogorov-Smirnov drift testing**, identifying feature-level prediction drivers and statistically significant shifts in sensor distributions.

WORK EXPERIENCE

Siemens | Software Engineer Intern | Remote Aug 2024 - Jul 2025

- Improved chatbot retrieval relevance by **40% through a Pinecone-backed RAG pipeline**, evaluating semantic-search quality across 1,200+ production conversations.
- Analyzed **AI quality and latency across 10,000 benchmark queries**, achieving 850 ms p95 response latency and identifying performance bottlenecks across retrieval and API stages.
- Led **3 interns through AI experimentation and feature development**, resolving 9 production issues through log analysis, debugging and service-level validation.

Vault of Codes | Software Engineering Intern | Remote Apr 2024 - Jun 2024

- Analyzed **checkout funnels and user-event data using SQL** across 8,000+ users and 1,600+ orders, identifying conversion bottlenecks in a marketplace generating $20K GMV.
- Increased checkout conversion from 41% to 50% (**22% relative improvement**) through **funnel analysis and one-click checkout optimization**, contributing $14K in revenue.
- Increased mobile conversion by **8% through performance analysis and iterative optimization**, reducing page load time to under 1.2 seconds.

Siddaganga Institute of Technology | Research Assistant | Tumkur May 2024 - Jan 2025

- Evaluated a **quantum-based 2Q learning approach for DDoS detection** across 5 datasets, achieving 95% detection performance.
- Improved model convergence by **30% through experimental optimization and comparative evaluation**, contributing to a peer-reviewed IEEE publication.

SKILLS

Languages: Python, SQL, Java, C++, TypeScript, JavaScript

Data Science & ML: Pandas, NumPy, Scikit-learn, LightGBM, XGBoost, PyTorch, TensorFlow, Optuna

Statistics & Experimentation: Regression, Classification, Time-Series Forecasting, Hypothesis Testing, A/B Testing, Feature Engineering, Model Evaluation, Threshold Optimization

Visualization & Explainability: Matplotlib, SHAP, LIME, Streamlit, Grafana

Data & MLOps: PostgreSQL, AWS S3, MLflow, Kafka, Redis, Docker, Git, GitHub Actions

CERTIFICATIONS & ACHIEVEMENTS

- Publication - Detection of DDoS Attacks in Fog Computing using Interpretable Quantum-based 2Q Learning Approach," IEEE. Jan 2025
- Open Source - Merged PR #472 into mindsdb/anton, fixing a bug that deleted saved user rules and adding 3 regression tests. Sep 2026